



Left Stream

1 Queries & Contract



Queries q_t



Contract $\alpha \in (0, 1)$
risk budget per step

2 Plan Catalog

Plan	Cost $c(p)$	Risk $r(p)$
p_1	c_1	r_1
p_2	c_2	r_2
$\vdots$	$\vdots$	$\vdots$
p_m	c_m	r_m



Decline $\perp$

$c_\perp$ 0

Costs: lower is better | Risks: in $[0, 1]$



Ledger-gated Controller

Ledger state B_t

1

Gate



Check budget

$$B_{t-1} \geq 1 - \alpha ?$$

Yes



Open
(optimise)

No



Closed
(force decline)

Action = Decline $\perp$
(risk 0, cost $c_\perp$)

2

Optimiser



Action LP (randomised)

$$\min_w \sum_{p \in P \cup \{\perp\}} w_p c(p)$$

s.t.

$$\sum_{p \in P \cup \{\perp\}} w_p \tilde{r}(p) \leq \alpha$$

$$w_p \geq 0, \sum w_p = 1$$

$$\leq 2 \text{ nonzero weights}$$

$\tilde{r}(p)$ optimistic risk estimate
(via estimates $\hat{r}_t$)

$$P = \{p_1, \dots, p_m\}$$

3

Accounting



Observe realised cost

$c_t \leftarrow c(\text{chosen action})$
(always observed)

Observe loss if audited

$\ell_t \in \{0, 1\}$
(1 if loss occurs, else 0)

Update ledger

$$B_t \leftarrow B_{t-1} + \alpha - \ell_t$$

Update estimates

$$\hat{r}_t \leftarrow \text{Update}(\hat{r}_{t-1}, \ell_t, p_t)$$



Right Service

Answer to Human



Answer

OR



Decline $\perp$

(cannot help)



Guarantees



DETERMINISTIC pathwise
 $V_t \leq \alpha t$ for all t
(NOT high probability)



No δ parameter



No $|P|$ -dependent price



Cost near OPT
(up to lower-order terms)

1

Ledger removes validity from estimates



Budget B_t tracks realised losses.

Safety is enforced pathwise by the ledger,
so risk estimates can be optimistic.

2

Decline unlocks α below the floor

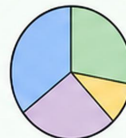


When $B_{t-1} < 1 - \alpha$, the gate closes
and forces Decline $\perp$ (risk 0).

This is what enables any α arbitrarily
close to 0 per step.

3

Randomised LP spends the allowance



The LP mixes over $P \cup \{\perp\}$ to satisfy
 $\sum w_p \tilde{r}(p) \leq \alpha$ while minimising cost.

At most two plans (plus possibly $\perp$)
receive nonzero weight.